

Interpretable PCOS Diagnosis using Clinical and Ultrasound Data

Abstract—Polycystic Ovary Syndrome (PCOS) is a prevalent endocrine disorder affecting women of reproductive age. Accurate diagnosis requires integrating clinical and imaging data, yet most computational approaches treat these modalities independently and lack interpretability.

This work proposes a dual-modality interpretable framework for PCOS diagnosis using decision-level fusion. Clinical data (541 patients) is processed using classical machine learning models (CatBoost, LightGBM, XGBoost) and transformer embeddings (BioClinicalBERT, BioBERT). Ultrasound images (11,784 samples) are analyzed using a fine-tuned Vision Transformer (ViT-B/16), achieving 99.6% accuracy.

Explainability is incorporated via SHAP for clinical features and Grad-CAM, Attention Rollout, and follicle segmentation for imaging. The fused system outputs a probability-based interpretable report combining clinical risk and imaging evidence, enabling actionable clinical decision support without requiring paired datasets.

Index Terms—Polycystic Ovary Syndrome, fusion model Learning, Vision Transformers, Clinical Data, Explainable AI, Medical Imaging

I. INTRODUCTION

Polycystic Ovary Syndrome (PCOS) is a complex hormonal disorder characterized by irregular menstrual cycles, hyperandrogenism, and polycystic ovarian morphology. The condition affects a significant percentage of women worldwide and is associated with long-term health risks such as infertility, type-2 diabetes, cardiovascular disease, and endometrial cancer. Conventional PCOS diagnosis relies on a combination of clinical assessment, biochemical tests, and pelvic ultrasound examination. However, manual interpretation of these modalities is time-consuming, subjective, and prone to inter-observer variability.

Recent advancements in machine learning (ML) and deep learning (DL) have enabled automated PCOS detection using clinical data or ultrasound images independently. Clinical datasets consisting of features such as age, body mass index (BMI), menstrual cycle length, hormone levels, and follicle counts have been used to train ML classifiers. Similarly, convolutional neural networks (CNNs) and, more recently, Vision Transformers (ViTs) have demonstrated promising performance in ultrasound image-based PCOS classification. Despite these advances, most existing studies focus on a single data modality.

In real-world clinical settings, diagnostic inputs are heterogeneous, and a robust decision-support system should be capable of handling either clinical data or imaging data. This

motivates the development of a multimodal PCOS detection framework that integrates knowledge from both domains while providing explainable predictions to support clinical trust.

II. LITERATURE REVIEW

Several studies have explored automated PCOS detection using machine learning and deep learning techniques.

A. PCOS Detection Using Clinical Data

Multiple works have applied classical ML algorithms such as Logistic Regression, Support Vector Machines (SVM), Random Forests, XGBoost, AdaBoost, and CatBoost on structured clinical datasets. These studies report high classification accuracy, with ensemble and gradient-boosting-based methods often outperforming traditional classifiers. Feature selection techniques have been shown to improve performance by identifying clinically relevant attributes such as BMI, menstrual irregularity, follicle count, and hyperandrogenic symptoms.

B. PCOS Detection Using Ultrasound Images

Ultrasound imaging is a key diagnostic tool for identifying polycystic ovarian morphology. Early approaches employed handcrafted feature extraction combined with ML classifiers. More recent studies use deep CNN architectures such as VGG, ResNet, Inception, and EfficientNet to automatically learn discriminative features from ovarian ultrasound images. These models achieve strong classification performance but often lack interpretability.

C. Vision Transformers in Medical Imaging

Vision Transformers (ViTs) have emerged as a powerful alternative to CNNs by modeling long-range dependencies through self-attention mechanisms. In PCOS detection, ViT-based models divide ultrasound images into patches, embed them into a latent space, and apply transformer encoders for classification. Attention maps generated by ViTs provide insights into diagnostically important regions, making them suitable for explainable medical imaging tasks.

D. Multimodal and Fusion-Based Approaches

Multimodal learning has gained attention in healthcare applications where heterogeneous data sources such as clinical records, medical images, and sensor data are available. Fusion strategies are typically categorized as early fusion, late fusion, or hybrid fusion. While multimodal frameworks have been successfully applied in diseases such as cancer and cardiovascular disorders, limited work exists in the context of PCOS, particularly involving transformers and explainability.

III. RESEARCH GAP

From the literature survey, the following research gaps are identified:

A. Unimodality

Most of the existing PCOS detection studies tend to rely on only one type of data, either clinical parameters (such as hormone levels, BMI, or menstrual history) or ultrasound images. Very few approaches attempt to combine both sources of information, even though clinicians typically use them together for accurate diagnosis. This separation limits the model's ability to capture the full clinical picture of PCOS.

B. Underexploration

Although transformer-based models have shown strong performance in many healthcare applications, their use for representing structured clinical data in PCOS diagnosis remains largely unexplored. Most current works still depend on traditional machine learning or basic deep learning methods, which may not effectively capture complex relationships among clinical features.

C. Heterogeneity

Most existing Vision Transformer-based approaches for PCOS diagnosis focus solely on ultrasound images and do not address the challenge of heterogeneous data integration. They fail to model clinical and imaging data through dedicated learning streams and lack effective multimodal fusion strategies, which limits their ability to reflect real-world diagnostic workflows.

D. Explainability

Model explainability in PCOS prediction systems is often treated as an afterthought. The majority of existing works rely on post-hoc explanation techniques such as feature importance or saliency maps. There is limited exploration of inherent attention-based interpretability, which could provide more transparent insights into how both clinical features and image regions contribute to the final diagnosis.

E. Inflexibility

Furthermore, there is a noticeable lack of flexible and modular diagnostic frameworks that can independently learn from heterogeneous input modalities while supporting effective multimodal fusion when available. Most existing systems assume the simultaneous availability of all data types, reducing robustness in real clinical scenarios where ultrasound images or certain clinical parameters may be missing or incomplete.

This project addresses these gaps by proposing a multimodal, transformer-based fusion framework for heterogeneous data that is capable of operating on clinical data or ultrasound images while providing explainable predictions.

IV. METHODOLOGY

For the implementation we used the below datasets:

A. Clinical Data Modeling

We utilize the Kaggle PCOS dataset (541 patients) involving 40 features. We benchmarked 11 classifiers and explored *BioClinicalBERT* and *BioBERT* embeddings by serializing clinical features into text strings. The dataset included features such as age, body mass index (BMI), menstrual cycle length, cycle regularity, weight gain, hirsutism, skin darkening, acne, dietary habits, and exercise patterns. The target variable indicated the presence or absence of PCOS.

Data preprocessing involved careful column standardization, type conversion, and handling of missing values. Categorical clinical indicators were converted into numerical representations, and inconsistencies in feature naming were resolved to ensure uniformity across samples. Missing values were imputed using median-based imputation to preserve the statistical distribution of the data while minimizing bias. Menstrual cycle regularity was explicitly re-encoded into a binary representation to improve interpretability and model learning.

The cleaned dataset was split into training and testing subsets using a stratified strategy to maintain class balance. An XGBoost classifier was employed due to its strong performance on structured tabular data and its ability to model non-linear feature interactions. The model was trained using gradient boosting with carefully selected hyperparameters, including a limited tree depth and a reduced learning rate to prevent overfitting.

The trained model achieved an overall classification accuracy of 86.24% on the test set. The confusion matrix analysis indicated strong performance in identifying non-PCOS cases, with a recall of 93%, while maintaining competitive sensitivity for PCOS-positive cases. Precision, recall, and F1-score metrics further confirmed the robustness of the model across both classes.

These results demonstrate the effectiveness of ensemble-based learning for clinical PCOS prediction and establish a reliable unimodal baseline for subsequent transformer-based clinical embedding and multimodal fusion.

B. Ultrasound Image Modeling

The imaging pipeline uses the **ViT-B/16** architecture [?] fine-tuned on 11,784 images from Figshare. We employ Attention Rollout and Grad-CAM for visual explainability, alongside an unsupervised follicle segmentation algorithm based on Otsu's thresholding. An ovarian ultrasound image dataset was used to train deep learning models for PCOS detection (Refer Fig. 2 and Fig. 3). Multiple CNN architectures were evaluated to identify the most suitable backbone for feature extraction.

After evaluating three different convolutional neural network architectures, a compact sequential CNN was found to deliver the best performance on the ultrasound image dataset. The model was designed to progressively capture meaningful local patterns from ovarian ultrasound images

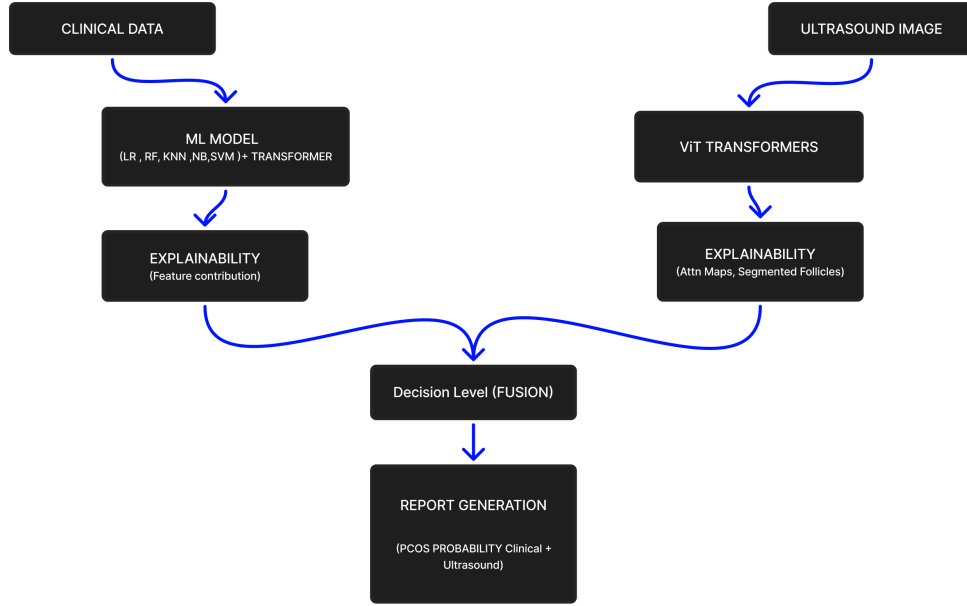


Fig. 1. Proposed dual-modality PCOS diagnosis architecture. Clinical and ultrasound branches operate independently, generate interpretable outputs, and are combined via decision-level fusion to produce the final diagnostic report.



Fig. 2. A sample ovarian ultrasound image showing polycystic morphology.



Fig. 3. A sample showing normal ovarian ultrasound.

through a series of convolution and pooling operations, while keeping the architecture relatively simple to avoid overfitting.

This model achieved a validation accuracy of 98.96% with a low loss value, indicating stable training and effective feature learning. Additional evaluation using precision, recall, F1-score, and confusion matrix analysis showed consistent performance across both PCOS and non-PCOS classes. These results demonstrate that carefully designed CNN-based models can effectively extract diagnostically relevant visual features from ultrasound images. The strong performance of this base-

line model provides a reliable reference point for subsequent experiments using Vision Transformers and attention-based explainability techniques.

C. System Overview

Fig. 1 illustrates the overall architecture of the proposed dual-modality PCOS diagnosis system. The pipeline consists of two independent branches—clinical data processing and ultrasound image analysis—followed by decision-level fusion and report generation.

In the clinical branch, structured patient data, including hormonal measurements, demographic features, and symptom indicators, is processed using a suite of machine learning models such as Logistic Regression (LR), Random Forest (RF), K-Nearest Neighbors (KNN), Naive Bayes (NB), and Support Vector Machine (SVM). Additionally, transformer-based embeddings are explored by converting tabular features into textual representations and encoding them using biomedical language models. The clinical pipeline outputs a probabilistic PCOS risk score p_c .

In parallel, the imaging branch processes ovarian ultrasound images using a Vision Transformer (ViT-B/16). (Refer Fig. 4, Fig. 5 and Fig. 6).

The model divides the input image into patches and applies multi-head self-attention to learn spatial patterns such as follicle distribution and ovarian morphology. This branch

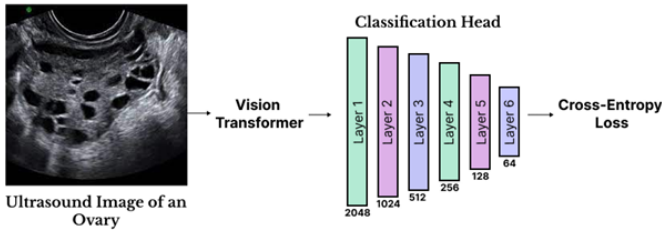


Fig. 4. The proposed framework for classification of Ultrasound images [2].

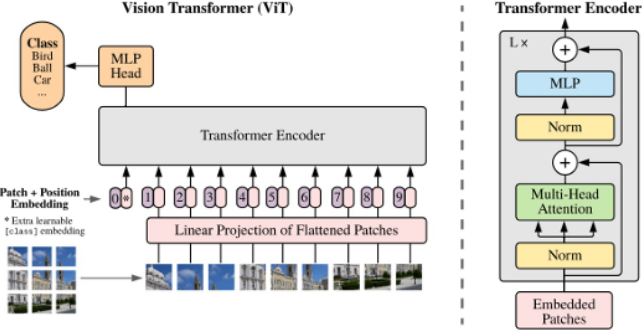


Fig. 5. Model overview, illustration of the Transformer was inspired by [3].

outputs a probability score p_i representing the likelihood of polycystic ovarian morphology (PCOM). Visual explainability is provided through attention maps, Grad-CAM visualizations, and follicle segmentation.

The outputs from both branches are combined using decision-level fusion, defined as:

$$p_{fused} = w_c \cdot p_c + w_i \cdot p_i, \quad (1)$$

where w_c and w_i represent modality weights. This strategy enables integration without requiring paired clinical and imaging datasets.

Finally, the system generates an interpretable report that includes the fused PCOS probability, key clinical feature contributions, visual evidence from ultrasound images, and follicle count estimates, providing actionable insights for clinical decision-making.

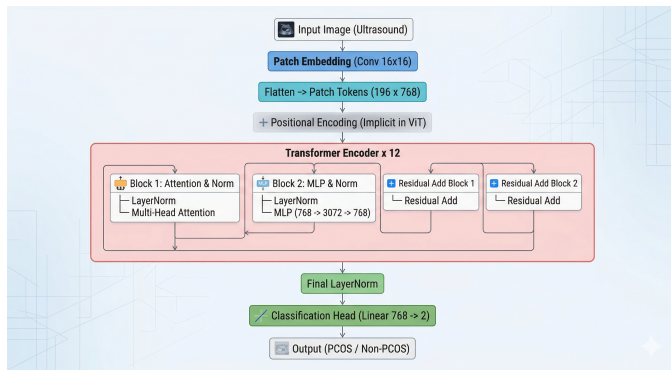


Fig. 6. VIT PIPELINE

Model	Accuracy	Precision	Recall	F1-score
LR	90.44	0.93	0.94	0.94
RF	88.23	0.96	0.88	0.92
KNN	88.97	0.91	0.94	0.93
NB	90.44	0.97	0.90	0.93
SVM	90.44	0.95	0.92	0.93
LGB	90.44	0.95	0.92	0.93
Adaboost	91.17	0.94	0.94	0.94
Catboost	93.38	0.95	0.96	0.96

Fig. 7. Base paper[1] Results

Ensemble (LR, CatBoost and AdaBoost)	Accuracy	Precision	Recall	F1-score
	88.23	0.96	0.88	0.92

Fig. 8. Base paper[1] Results-Ensemble Methods

Establishing these unimodal baselines serves two key purposes: first, it enables a fair performance comparison with more advanced transformer-based models planned in later stages of the project; second, it provides a solid foundation for analyzing the benefits of attention mechanisms, long-range feature modeling, and multimodal fusion in improving diagnostic accuracy and interpretability.

D. Evaluation Metrics

Both clinical and image-based models were evaluated using standard metrics including accuracy, precision, recall, F1-score, and confusion matrices. These results establish strong unimodal baselines for subsequent multimodal fusion.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	93.5780	0.8718	0.9444	0.9067
KNN (k=10)	89.9083	0.9630	0.7222	0.8254
Naive Bayes	88.0734	0.7805	0.8889	0.8312
SVM	91.7431	0.9091	0.8333	0.8696
Decision Tree (Gini)	81.6514	0.7500	0.6667	0.7059
Decision Tree (Entropy)	81.6514	0.7353	0.6944	0.7143
Random Forest	91.7431	0.9355	0.8056	0.8657
LGB	94.4954	0.9167	0.9167	0.9167
AdaBoost	93.5780	0.9394	0.8611	0.8986
CatBoost	95.4128	0.9429	0.9167	0.9296
XGBoost	94.4954	0.9412	0.8889	0.9143
Ensemble (LR, AdaBoost, CatBoost)	92.6606	0.9375	0.8333	0.8824

Fig. 9. BasePaper[1] Our implementation Results

Model	Baseline Accuracy	Transformer Model	Transformer Accuracy	Transformer Precision	Transformer Recall	Transformer F1-score	Accuracy Gain/Loss
AdaBoost	93.5780	Bio_ClinicalBERT	81.6514	0.8333	0.5556	0.6667	-11.9266
AdaBoost	93.5780	bert-base-uncased	79.8165	0.7917	0.5278	0.6333	-13.7615
AdaBoost	93.5780	biobert-base-cased-v1.2	77.9817	0.7727	0.4722	0.5862	-15.5963
AdaBoost	93.5780	BiomedNLP-PubMed BERT-base-uncased-a bstract-ful...	75.2294	0.6800	0.4722	0.5574	-18.3486

Fig. 10. Adaboost on Transformer embeddings

CatBoost	95.4128	Bio_ClinicalBERT	84.4037	0.8519	0.6389	0.7302	-11.0092
CatBoost	95.4128	bert-base-uncased	83.4862	0.8462	0.6111	0.7097	-11.9266
CatBoost	95.4128	biobert-base-cased-v1.2	83.4862	0.8462	0.6111	0.7097	-11.9266
CatBoost	95.4128	BiomedNLP-PubMed BERT-base-uncased-a bstract-ful...	77.9817	0.8000	0.4444	0.5714	-17.4312

Fig. 11. CatBoost on transformer embeddings

LightGBM	94.4954	Bio_ClinicalBERT	86.2385	0.8621	0.6944	0.7692	-8.2569
LightGBM	94.4954	biobert-base-cased-v1.2	83.4862	0.8750	0.5833	0.7000	-11.0092
LightGBM	94.4954	bert-base-uncased	80.7339	0.7778	0.5833	0.6667	-13.7615
LightGBM	94.4954	BiomedNLP-PubMed BERT-base-uncased-a bstract-ful...	77.0642	0.7895	0.4167	0.5455	-17.4312

Fig. 12. Light GB - Transformers results

Logistic Regression	93.5780	Bio_ClinicalBERT	79.8165	0.6842	0.7222	0.7027	-13.7615
Logistic Regression	93.5780	bert-base-uncased	78.8991	0.6970	0.6389	0.6667	-14.6789
Logistic Regression	93.5780	BiomedNLP-PubMed BERT-base-uncased-a bstract-ful...	75.2294	0.6286	0.6111	0.6197	-18.3486
Logistic Regression	93.5780	biobert-base-cased-v1.2	75.2294	0.6216	0.6389	0.6301	-18.3486

Fig. 13. Logistic Regression-Transformers results

Random Forest	91.7431	Bio_ClinicalBERT	77.0642	0.8235	0.3889	0.5283	-14.6789
Random Forest	91.7431	bert-base-uncased	73.3945	0.8889	0.2222	0.3556	-18.3486
Random Forest	91.7431	BiomedNLP-PubMed BERT-base-uncased-a bstract-ful...	71.5596	1.0000	0.1389	0.2439	-20.1835
Random Forest	91.7431	biobert-base-cased-v1.2	71.5596	0.7273	0.2222	0.3404	-20.1835

Fig. 14. Random Forest - Transformers results

XGBoost	94.4954	bert-base-uncased	83.4862	0.8462	0.6111	0.7097	-11.0092
XGBoost	94.4954	Bio_ClinicalBERT	82.5688	0.8148	0.6111	0.6984	-11.9266
XGBoost	94.4954	biobert-base-cased-v1.2	82.5688	0.8400	0.5833	0.6885	-11.9266
XGBoost	94.4954	BiomedNLP-PubMed BERT-base-uncased-a bstract-ful...	75.2294	0.7368	0.3889	0.5091	-19.2661

Fig. 15. XGBoost-Transformers results

===== TEST RESULTS =====

Accuracy : 0.9960407239819005
Precision: 0.9960773355170746
Recall : 0.9960407239819005
F1 Score : 0.9960430896784005

Confusion Matrix

```
[[1011  7]
 [  0 750]]
```

Total Misclassified Images: 7

Fig. 16. Metrics and Confusion matrix on test dataset

V. RESULTS

A. Clinical Dataset (Baseline Implementation)

The performance comparison of different machine learning models on the clinical dataset is shown in Fig.7 and Fig.8. Among all the models, CatBoost achieved the highest accuracy of 93.38%, along with strong precision, recall, and F1-score, making it the best-performing model. Models like Logistic Regression, SVM, and LightGBM also showed stable performance with accuracy around 90

Ensemble methods were also tested; however, the combined model (LR, CatBoost, and AdaBoost) did not outperform the best individual model, indicating that not all ensemble combinations necessarily improve results.

Overall, the results show that tree-based and boosting models perform better on clinical data, as they can capture complex relationships between features. These findings provide a solid baseline for further improvements using advanced models in the later sections.

B. Our Implementation (Clinical Dataset)

After implementing the base paper approach on our dataset, we obtained the results shown in Fig. 9. It can be observed that CatBoost achieved the highest accuracy of 95.41%, outperforming all other models. LightGBM and XGBoost also showed strong performance with accuracies around 94.5%, indicating that boosting-based methods are highly effective for this task.

Compared to the baseline results, most models in our implementation showed improved performance, suggesting better preprocessing and model tuning. However, the ensemble model did not outperform the best individual model, which indicates that combining models does not always guarantee improvement.

Overall, the results confirm that tree-based boosting models are the most suitable for clinical PCOS prediction, and our implementation achieves better performance than the base paper, providing a stronger foundation for further analysis.

C. Clinical Data using Transformer Embeddings

In this experiment, instead of using raw tabular data directly, we converted the clinical features into a textual format and extracted embeddings using transformer models such as BioClinicalBERT, BioBERT, BERT, and PubMedBERT. These embeddings were then used as input features for traditional machine learning models like Logistic Regression, Random Forest, LightGBM, and XGBoost.

From the results in Fig.10, 11, 12, 13, 14, it can be observed that transformer-based embeddings did not consistently outperform the original tabular approach. While some combinations, such as LightGBM with BioClinicalBERT and XGBoost with BERT embeddings, showed reasonable performance, the overall accuracy and F1-scores were generally lower compared to using raw clinical features.

This suggests that, for structured clinical data, directly using tabular features is more effective than converting them into text and applying transformer embeddings. However, this approach still provides useful insights and demonstrates an alternative way of leveraging pre-trained language models for medical data.

D. Vision Dataset(BaseLine Implementation)

In this section, we present the results (Fig. 16) obtained by implementing the second base paper, which focuses on image-based PCOS detection using deep learning techniques. Ultrasound images were used as input, and a Vision Transformer (ViT) model was trained to classify images into infected and non-infected categories.

Experimented with the learning rate values: $1e-3$, $1e-4$, $3e-5$, $1e-5$, and $3e-5$ performed well with best validation accuracy 99.437%

The model was trained using a train-validation-test split, and its performance was evaluated using standard metrics such as accuracy and F1-score. The results show that the ViT model is able to effectively capture important visual patterns in ultrasound images, particularly the presence and distribution of follicles. The misclassification results are given in Fig. 17.

E. Explainability and Segmentation Pipeline

In this work, ultrasound images are first processed through a Vision Transformer (ViT), where attention mechanisms are leveraged to capture global contextual relationships. Attention rollout is performed to visualize the contribution of different regions in the image toward the model's prediction.

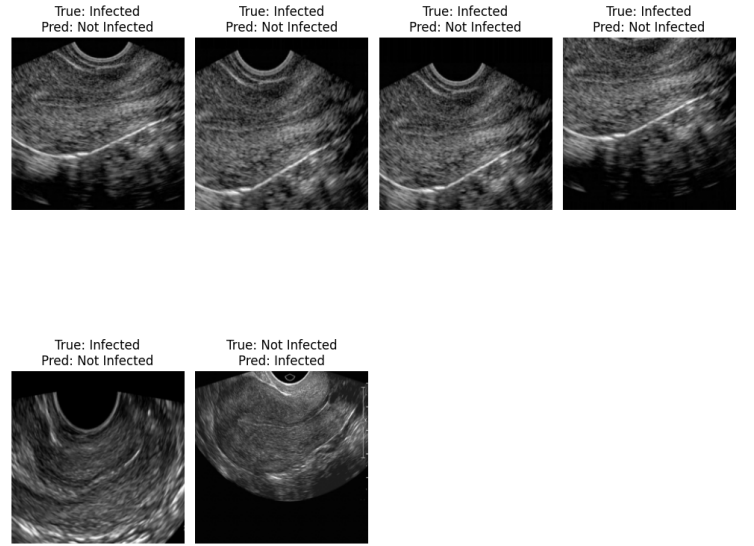


Fig. 17. Misclassified images by Vision Transformer on PCOS Ultrasound Dataset

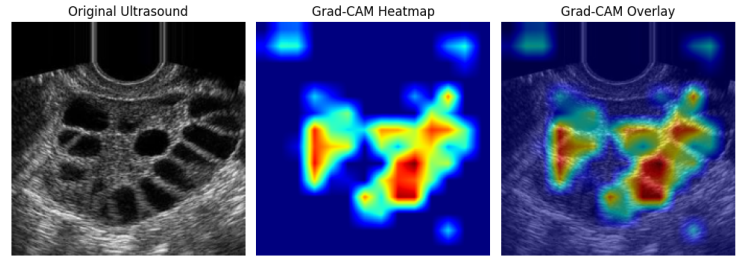


Fig. 18. Grad-Cam Attention Heatmap for Image1.

Subsequently, Gradient-weighted Class Activation Mapping (Grad-CAM) is applied to generate heatmaps, highlighting the most relevant regions influencing the model's decision. These heatmaps provide coarse localization of important structures within the ultrasound images.

Initially, direct application of MedSAM (Medical Segment Anything Model) did not yield satisfactory segmentation results. To overcome this limitation, Grad-CAM heatmaps are utilized as guiding priors. Specifically, the Grad-CAM outputs are used as overlays and prompts to condition MedSAM, enabling it to focus on clinically relevant regions. This guidance significantly improves the quality of follicle segmentation. (Refer Fig.18 - Fig.23).

In addition to the learning-based pipeline, a classical image processing approach is also implemented independently. This method performs segmentation without relying on Grad-CAM or deep learning-based attention mechanisms, serving as a baseline for comparison. (Refer Fig.24 - Fig.28).

Overall, the proposed pipeline integrates transformer-based feature extraction, Grad-CAM-based localization, and MedSAM-guided segmentation, alongside a traditional image processing method for benchmarking.

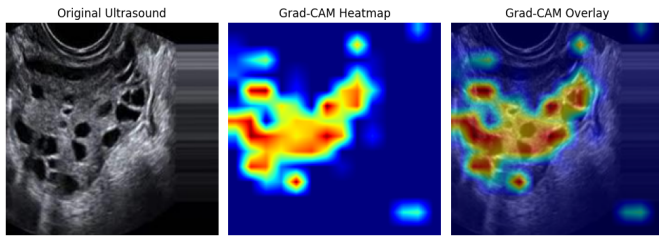


Fig. 19. Grad-Cam Attention Heatmap for Image2.

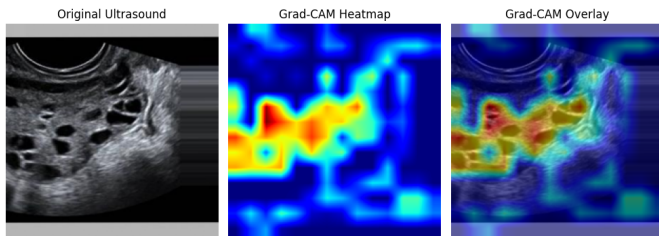


Fig. 20. Grad-Cam Attention Heatmap for Image3.

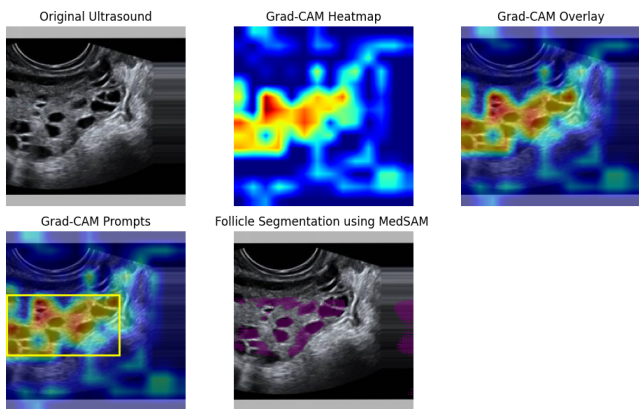


Fig. 21. Segmentation using GradCAM + MedSAM for image1.

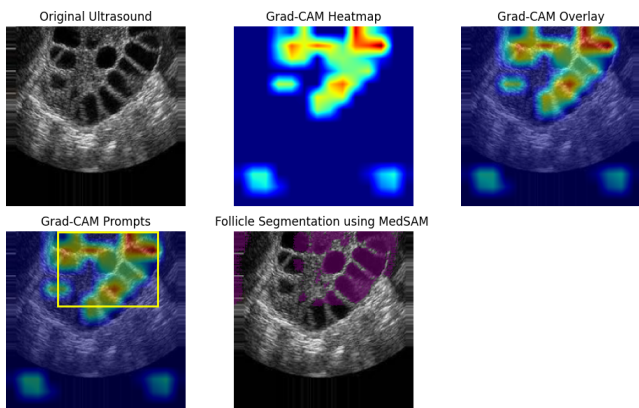


Fig. 22. Segmentation using GradCAM + MedSAM for image2

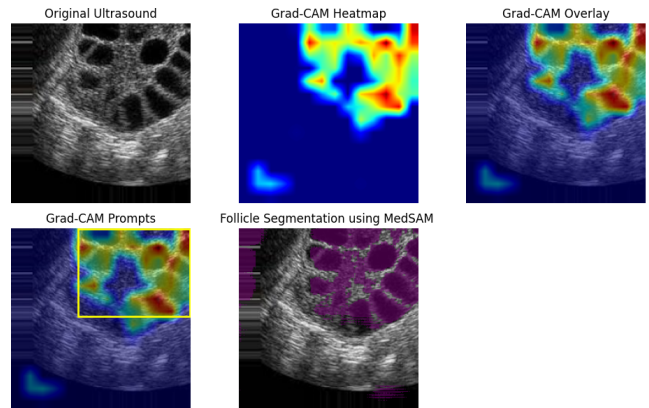


Fig. 23. Segmentation using GradCAM + MedSAM for image3



Fig. 24. Segmentation using Classical Iage Processing Technique for image1.



Fig. 25. Segmentation using Classical Iage Processing Technique for image2.



Fig. 26. Segmentation using Classical Iage Processing Technique for image3.

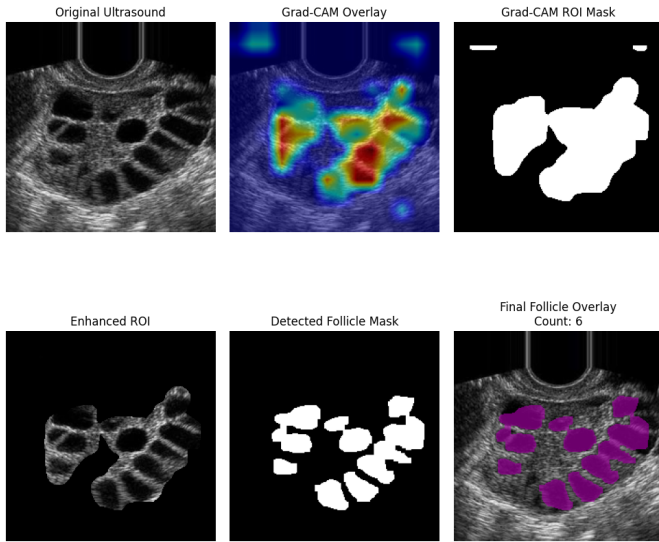


Fig. 27. Segmentation using Grad-Cam + Classical Iage Processing Technique for image1.

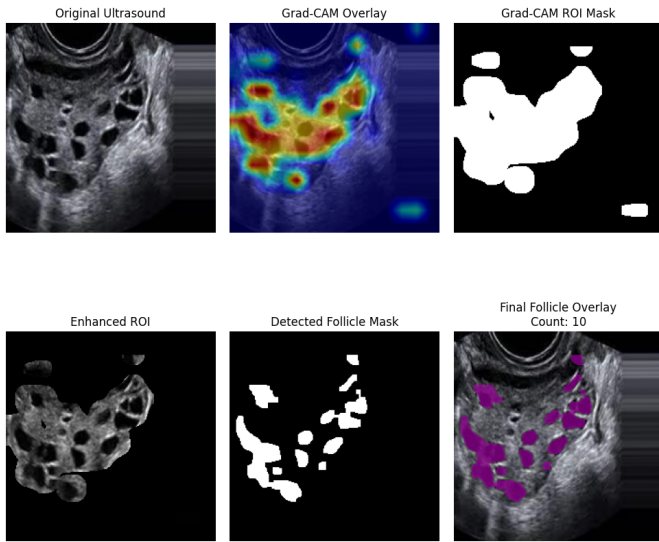


Fig. 28. Segmentation using Grad-Cam + Classical Iage Processing Technique for image2.

VI. CONCLUSION

This report presented the motivation, literature survey, identified research gaps, and current progress towards a explainable PCOS detection. By leveraging transformers for clinical data and Vision Transformers for ultrasound images, the proposed approach aims to provide accurate, flexible, and interpretable PCOS diagnosis. The future work will focus on implementing and validating the fusion model, with the ultimate goal of contributing a novel, explainable multimodal solution to PCOS diagnosis.

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